

Team Fosuai at PAN 2026: Source Retrieval via Query-Chunk Provenance Modeling for Generative Plagiarism Detection

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Abstract

The PAN@CLEF 2026 Generative Plagiarism Detection task aims to retrieve potential source documents for a given suspicious document from a large-scale source corpus. Addressing the problem that texts generated by large language models often fuse content from multiple sources, diluting the signal of full-document-level queries, this paper proposes a lightweight source retrieval method based on query chunk provenance modeling. The method splits the suspicious document into paragraph-level query chunks, performs BM25 retrieval independently for each chunk, and aggregates candidate sources via a coverage-weighted voting mechanism, thereby strengthening signals from sources consistently matched across multiple chunks. The system relies only on traditional inverted indexes and standard Python libraries, requiring no GPU or pre-trained language models, and has been submitted as the fosuai team's TIRA entry for the official PAN26 evaluation. Experimental and official results show that the method achieves competitive early-rank performance, while also revealing limitations in output depth and cross-dataset generalization that require further improvement.

Keywords

Generative Plagiarism Detection, Source Retrieval, Query Segmentation, Provenance Modeling, BM25, Coverage-Weighted Voting

1. Introduction

The PAN@CLEF 2026 Generative Plagiarism Detection task addresses a rapidly emerging challenge: Large Language Models (LLMs) can synthesize, recombine, and rewrite content from dozens of source documents into a single suspicious text, where different paragraphs may originate from entirely different sources. The task consists of two stages: Source Retrieval—identify source documents from a large corpus given a suspicious document; and Text Alignment—precisely locate plagiarized passages at character level. This paper focuses on Stage 1.

Existing source retrieval methods predominantly adopt a document-level querying paradigm: the entire suspicious document is encoded as a single query vector or bag-of-words, and a relevance score is computed against each candidate document. While effective for single-source plagiarism, this paradigm suffers from multi-source signal dilution. When a suspicious text composites content from Source A's introduction, Source B's methodology, and Source C's results, the query signal becomes a noisy mixture of multiple provenance fingerprints. Each individual source's matching signal is diluted by noise from other sources, causing the true source to be ranked outside the top retrieval positions.

To address this problem, we propose Query-Chunk Provenance Modeling. Rather than treating the suspicious document as a monolithic query, we decompose it into paragraph-level semantic segments. Each segment independently retrieves candidate sources, and a coverage-weighted voting mechanism aggregates the results. Documents matched by multiple independent segments receive a coverage bonus—a strong signal of genuine multi-paragraph provenance. This approach explicitly models the provenance structure of multi-source plagiarism while maintaining a lightweight implementation.

2. Related Work

Source retrieval has been a core component of PAN plagiarism detection tasks since PAN 2014 错误!未找到引用源。 . Early approaches¹ relied on lexical matching: word n-gram overlap, fingerprinting, and BM25 [3] have proven effective for verbatim and lightly-paraphrased plagiarism. Dense retrieval methods using sentence

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encoders such as Sentence-BERT [2] and E5 [3] project documents into a dense vector space for semantic matching, achieving better generalization on heavily paraphrased texts. Chunk-based retrieval [4] splits documents into fixed-size segments for independent encoding and max-pooling aggregation, but applies splitting only on the document side while keeping the query as a single unit. Query decomposition techniques [5] split complex queries into sub-queries for multi-hop QA, but have not been applied to plagiarism-specific segmentation strategies. Our work differs from all of the above by modeling provenance structure on the query side: decomposing the suspicious document itself to isolate individual source signals before retrieval.

3. Method

3.1. Task Formulation

Given a source corpus $C = \{d_1, d_2, \dots, d_N\}$ ($N = 60,592$ academic documents from ClueWeb09) and a suspicious document q , source retrieval aims to produce a ranked list $R(q) = [d_{r_1}, \dots, d_{r_K}]$ such that the true source document $d^*(q)$ appears as high as possible. In generative plagiarism, q may be synthesized from multiple sources, though evaluation considers one primary source per q . Retrieval quality is measured by Recall@K, nDCG@K, MRR, and Reciprocal Rank (RR).

3.2. Query-Chunk Provenance Modeling

Our method consists of three stages.

Overall architecture. Suspicious document $\rightarrow$ Paragraph split $\rightarrow$ Query chunks $C_1..C_n$. Center: Each chunk independently performs BM25 retrieval $\rightarrow$ Segment-level candidate lists. Right: Coverage-weighted voting $\rightarrow$ Final ranked source list. Dashed boxes highlight two key modules: Paragraph Segmentation and Coverage-Weighted Voting.

Stage 1—Paragraph-Level Query Segmentation. The suspicious document q is split by double-newline paragraph boundaries. Adjacent paragraphs are merged until reaching `max_chars=3,000`, with chunks shorter than `min_chars=800` discarded. Output is capped at 20 chunks. This produces semantically coherent segments roughly corresponding to academic document section structure. For documents lacking paragraph structure, the entire text is treated as a single segment.

Stage 2—Independent Segment Retrieval. Each segment c_i independently performs BM25 retrieval with optimized parameters: `max_terms=20` (vs. 100 for full-document queries, since short segments contain fewer discriminative terms) and `max_df=5,000` (filtering terms appearing in $>5,000$ documents, $\sim 8\%$ of the corpus). Without filtering, segment-level BM25 is 85–150 $\times$ slower than full-document BM25 because segments contain mostly common academic function words with very long posting lists (30K–50K entries). The `max_df` threshold eliminates these non-discriminative terms, achieving an $\sim 20\times$ speedup (1.7–3.0s $\rightarrow$ 0.15s per segment). Heap-based top-K extraction (`heapq.nlargest`) replaces full sorting for additional efficiency.

Stage 3—Coverage-Weighted Source Voting. Segment-level rankings are aggregated into a final ranking via: $\text{Score}(d) = \sum_i \text{BM25}(d|c_i) \times (1 + \text{count}(d)/n)$, where $\text{count}(d)$ is the number of segments that retrieved document d in their top-50 results, and n is the total number of segments. The coverage bonus $(1 + \text{count}(d)/n) \in [1, 2]$ rewards documents consistently matched by multiple independent segments—a strong signal of genuine multi-paragraph provenance rather than incidental single-phrase matching. This mechanism can be viewed as constructing an explicit provenance graph: nodes are query segments and candidate documents, edges represent segment-document matches, and the final ranking reflects each document's global importance in the graph.

4. Experiments

4.1. Experimental Setup

4.1.1. Dataset

Development data comes from the PAN 2025 generative plagiarism detection dataset converted to PAN26 format: 60,592 source documents from ClueWeb09 (avg. 11,765 tokens), 42,444 LLM-generated suspicious documents (training), 5,522 held-out queries with 7,949 source documents, and 43,140 query-document relevance annotations (qrels). The PAN26 official spot-check dataset (4 queries) was used only for format verification and TIRA submission validation. Official main test set results are from TIRA (accessed May 25, 2026).

4.1.2. Preprocessing

All texts are lowercased and tokenized via regex `[a-z0-9]+`. An inverted index is built with 1,273,004 unique terms. BM25 parameters: $k_1=1.2$, $b=0.75$. Paragraph segmentation: `min_chars=800`, `max_chars=3,000`. Segment retrieval: `max_terms=20`, `max_df=5,000`, `top_k=50`. Standard retrieval: `max_terms=100`, `top_k=100`. All metrics are evaluated using `scripts/evaluate.py` (supporting multi-relevant qrels).

4.1.3. Evaluation Metrics

Four standard IR metrics are used: (1) Recall@K (K=10, 100)—proportion of queries where the correct source appears in the top-K results; (2) nDCG@10—Normalized Discounted Cumulative Gain with position weighting; (3) MRR—Mean Reciprocal Rank; (4) RR—Reciprocal Rank, used by the official TIRA evaluation. All metrics range [0, 1], higher is better.

4.1.4. Baselines

Three comparison categories: (1) Standard BM25—full document as single query, `max_terms=100`. (2) E5 Chunking—256-token document chunks encoded with E5-base-v2, coverage-aware scoring for BM25 blind-spot queries. (3) Ablation variants—fixed-chunk segmentation (3,000-char rolling windows) and semantic segmentation (section header pattern matching). All methods share identical inverted indexes and BM25 parameters for fair comparison.

4.2. Results and Analysis

Table 1

Development data method overview

Method	R@10	R@100	nDCG@10	MRR	Data Scale
Standard BM25	0.9196	0.9769	0.8184	0.7856	42,444q (full train)
+ Query Decomposition	0.9317	0.9890	0.8274	0.7937	982q (Cat1 blind-spot)
+ E5 Chunking	0.9830	0.9917	0.8716	0.8355	42,444q (full train)
Segmented BM25 (ours)*	0.9700	0.9912	0.9242	0.9092	800q sample (TIRA code)

*Segmented BM25 result from 800-query random sample (seed=20260520) using submit_pan26.py (exact TIRA submission code). Query Decomposition recovered 514/982 Cat1 blind-spot queries (52.3% recovery rate). E5 Chunking attains higher R@10 but requires a 440MB E5 model; our method is numpy-only.

Table 2

Held-out validation (7,949 documents, 5,522 queries)

Method	R@10	R@100	nDCG@10	MRR
Standard BM25	0.9804	0.9874	0.9323	0.9174
Segmented BM25 (ours)	0.9947	0.9969	0.9770	0.9724

Table 3

Segmentation strategy comparison (held-out)

Strategy	R@10	MRR	Segmented%	Time (s)
No segmentation (BM25)	0.9804	0.9174	–	33
Paragraph split (ours)	0.9947	0.9724	99.9%	358
Fixed chunk	0.9933	0.9705	99.9%	300
Semantic split (section headers)	0.9746	0.9195	51.6%	49

Table 3 reveals several findings: (1) Paragraph segmentation achieves the best overall performance. (2) Fixed-chunk segmentation performs close to paragraph split but is slightly inferior, confirming that paragraph boundaries provide more meaningful semantic units than arbitrary character windows. (3) Semantic segmentation based on section header pattern matching (Introduction, Method, Results, etc.) actually degrades performance below standard BM25 (R@10 0.9746 vs. 0.9804), because only 51.6% of queries contain explicit section headers. This negative result is retained as a credibility-enhancing finding: simple rule-based semantic segmentation is not viable for this dataset.

Table 4

Ablation study—speed optimization (200 queries)

Configuration	R@10	nDCG@10	MRR	Time/chunk
(a) Standard BM25	0.900	0.805	0.775	~0.02s
(b) + segmentation (max_terms=50)	0.975	0.942	0.931	1.7–3.0s
(c) + max_df=5,000	0.973	0.940	0.929	~0.30s
(d) + max_terms=20 + heapq	0.975	0.942	0.931	~0.15s

The ablation study demonstrates that the three optimization strategies (max_df filtering, max_terms reduction, heapq top-K) together achieve an $\sim 15\text{--}20\times$ speedup with negligible accuracy loss ($\Delta R@10 = 0.002$, within statistical noise).

Table 5
 Official PAN26 main test results (TIRA, accessed May 25, 2026)

System	Recall@10	Recall@100	nDCG@10	nDCG@100	RR
BM25 baseline (official)	0.5767	0.7825	0.5532	0.6068	0.8318
fosuai / oscillating-list (ours)	0.6300	0.6300	0.6263	0.6263	0.9350
JLP / jlp (best system)	0.8542	0.9629	0.7994	0.8326	0.9760

On the official main test, our system achieves $RR=0.9350$ (ranked 3rd), demonstrating strong early-rank performance consistent with the design goal of coverage-weighted voting. Recall@10 and nDCG@10 both improve over the BM25 baseline (+9.2% and +13.2% respectively). However, Recall@100 equals Recall@10 (0.6300) because the submitted run file contained only top-10 documents per query rather than the expected top-1,000. This is an engineering limitation to be corrected in future submissions. The gap between development-set $R@10$ ($\sim 0.97\text{--}0.99$) and test-set $R@10$ (0.63) reflects differences in corpus size, domain, and plagiarism type distribution, underscoring that cross-dataset generalization remains the core challenge of this task.

5. Conclusions

We presented Query-Chunk Provenance Modeling, a lightweight (numpy-only) source retrieval method for PAN26 generative plagiarism detection. The method decomposes suspicious documents into paragraph-level segments, independently retrieves sources per segment via BM25, and aggregates results through coverage-weighted voting to construct an explicit provenance graph. On development data, our method consistently outperforms standard BM25. On the official PAN26 main test, our system (fosuai / oscillating-list) achieved $RR=0.9350$ (3rd by RR). The submitted run's top-10 output limit constrained Recall@100; future work should output top-1000 and explore topic-based or embedding-based segmentation strategies for improved semantic coherence. PAN26 official secondary dataset results are pending.

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Declaration on Generative AI

During the preparation of this work, the author(s) used Claude in order to: English abstract translation, and English language proofreading. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

6. References

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